

Lecture Notes: Fourier Representation of Signals and LTI Systems

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1 Introduction

Fourier analysis allows us to represent a signal as a weighted sum of complex sinusoids. This approach is essential for analyzing Linear Time-Invariant (LTI) systems, particularly in understanding frequency response and system behavior with sinusoidal inputs.

Definition: Complex Sinusoid - A complex sinusoid is a signal of the form $x(t) = e^{j\omega t}$, where ω denotes angular frequency. Complex sinusoids are the fundamental building blocks in Fourier analysis due to their invariance property when passed through LTI systems.

2 Complex Sinusoids and LTI Systems

The response of an LTI system to a complex sinusoid input $x[n] = e^{j\Omega n}$ is given by the system's frequency response $H(e^{j\Omega})$, scaled by $e^{j\Omega n}$. The output $y[n]$ becomes:

$$y[n] = H(e^{j\Omega}) \cdot e^{j\Omega n}$$

A table below contrasts continuous and discrete frequency responses, explaining similarities and differences.

	Continuous-Time	Discrete-Time
Input Signal	$x(t) = e^{j\omega t}$	$x[n] = e^{j\Omega n}$
Frequency Response	$H(j\omega) = \int_{-\infty}^{\infty} h(t)e^{-j\omega t} dt$	$H(e^{j\Omega}) = \sum_{k=-\infty}^{\infty} h[k]e^{-j\Omega k}$
Output	$y(t) = H(j\omega)e^{j\omega t}$	$y[n] = H(e^{j\Omega})e^{j\Omega n}$

Table 1: Comparison of Continuous and Discrete Frequency Responses

3 Four Classes of Signals

Fourier transforms provide different representations depending on the signal's periodicity and time type. The table below summarizes Fourier Series, Fourier Transform, DTFS, and DTFT.

	Periodic	Non-Periodic
Continuous-Time	Fourier Series (FS)	Fourier Transform (FT)
Discrete-Time	Discrete-Time Fourier Series (DTFS)	Discrete-Time Fourier Transform (DTFT)

Table 2: Four Classes of Fourier Representations

4 Important Terms

Key terms in Fourier analysis include frequency response, complex sinusoid, Fourier Transform, impulse response, and phase response. The table below defines each term for clarity.

Term	Definition
Frequency Response	The response of an LTI system to sinusoidal inputs, represented by $H(e^{j\Omega})$ or $H(j\omega)$.
Complex Sinusoid	A sinusoidal signal expressed as $e^{j\omega t}$ or $e^{j\Omega n}$, fundamental in frequency domain analysis.
Fourier Transform	Converts a time-domain signal into its frequency components.
Impulse Response	The output of an LTI system when the input is a unit impulse.
Phase Response	The phase shift introduced by an LTI system, often derived from $\arg(H(j\omega))$.

Table 3: Definitions of Important Terms

5 Examples and Proofs

5.1 Mathematical Proofs

Consider the RC Circuit with an impulse response $h(t) = \frac{1}{RC}e^{-\frac{t}{RC}}u(t)$. The frequency response is derived by taking the Fourier transform:

$$H(j\omega) = \int_0^{\infty} \frac{1}{RC} e^{-\frac{t}{RC}} e^{-j\omega t} dt$$

Solving, we get:

$$H(j\omega) = \frac{1}{1 + j\omega RC}$$

5.2 MATLAB Example

Using MATLAB, we can plot the magnitude and phase response of the RC circuit.

```
1 R = 1; C = 1;
2 omega = linspace(-10, 10, 1000);
3 H = 1 ./ (1 + 1j*omega*R*C);
4
5 % Magnitude and Phase
6 magnitude = abs(H);
7 phase = angle(H);
8
9 % Plotting
10 subplot(2,1,1);
11 plot(omega, magnitude);
12 title('Magnitude Response');
13 xlabel('Frequency (rad/s)');
14 ylabel('|H(j\omega)|');
15
16 subplot(2,1,2);
17 plot(omega, phase);
18 title('Phase Response');
19 xlabel('Frequency (rad/s)');
20 ylabel('H (j\omega)');
```

Listing 1: MATLAB Code for RC Circuit Frequency Response

6 Applications of Fourier Representation

Fourier Series and Transform find applications in signal filtering and analysis, crucial for engineering systems where frequency-specific modifications are required.